

Audio-driven Talking Head Generation with Transformer and 3D Morphable Model

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ABSTRACT

In the task of talking head generation, it is hard to learn the mapping relationship between generated head image and input audio signal. To tackle this challenge, we propose to learn the mapping relationship between input audio signal and the parameters of three-dimensional morphable face model (3DMM) first, which is easier to learn. Then the parameters of 3DMM are used to guide the generation of high-quality talking head images. Prior works mostly encode audio features from short audio windows, which may influence the accuracy of lip movements sometimes because of the limited context. In this paper, we propose a transformer-based audio encoder to take full use of the long-term context from audio and then predict a sequence of 3DMM parameters accurately. Unlike prior works that only use the 3DMM parameters of expression, rotation and translation, we propose to include the parameters of identity. Since the location of 3D facial mesh point is decided by the expression and identity parameters, it is helpful to supply more subtle control of lip movement by considering the identity parameters. The experimental results reveal that our method ranks first in 4 of the total 11 evaluation metrics, which ranks first in the talking head generation track.

CCS CONCEPTS

• Computing methodologies → Computer vision; Phonology / morphology.

KEYWORDS

talking head generation, audio-driven, transformer, long-term context, 3D morphable model

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1 INTRODUCTION

The task of talking head generation aims to generate a video in which the speaker's lip movement is consistent with the input audio. It has great application potential in the field of virtual reality, digital human, teleconference, film production, etc. Some works [10, 20, 23] try to generate the talking head of a target speaker by using driving signal from a source speaker without the need of input audio. In other works [9, 14, 17, 34], the input audio is used to drive the lip movement of target speaker directly since the strong correlation between them. However, the correlation between lip movement and audio is hard to learn since it is a cross-modal task with image and audio from visual and auditory modalities separately.

A conversational head generation challenge is held in ACM Multimedia 2022 to encourage the participants to solve the challenge of vivid talking head video generation and responsive listening head video generation. This challenge is based on the ViCo dataset [33]. The ViCo dataset not only contains the videos of different speakers but also contains the corresponding videos of listeners. There are a total number of 92 identities and 483 paired "speaking-listening" video clips in the ViCo dataset including 67 speakers and 76 listeners. In this challenge, we focus on the track of vivid talking head video generation conditioned on the identity and audio signals of the speaker.

To deal with the above challenge of talking head generation, we propose to use the parameters of three-dimensional morphable model (3DMM) as intermediate signal between audio and image. Specifically, we first learn the mapping relationship between audio signal and parameters of 3DMM. Then the parameters are used as the driving signal for generating the photo-realistic talking head image. Prior works [6, 32] mostly extract the audio feature corresponding to one video frame from an audio window which only contains short context. However, we argue that the long-term context is important for the accuracy and continuity of generated lip movement. In this paper, we propose a transformer-based audio encoder to extract the long-term audio context by the weighted combination of audio sequences and map it to the parameters of 3DMM. The parameters are disentangled into several parts: expression, identity, rotation, translation, etc. The prior works tend to ignore the parameters of identity. However, when deciding the location of 3D facial mesh points, the expression and identity parameters are both needed, which means that expression and identity parameters are both highly related to the lip movement. What's more, the identity parameters can help model learn each speaker's own speaking style, which is helpful for generating higher quality images. So we propose to incorporate identity parameters for model to learn accurate lip movement implicitly and generate high-quality images.

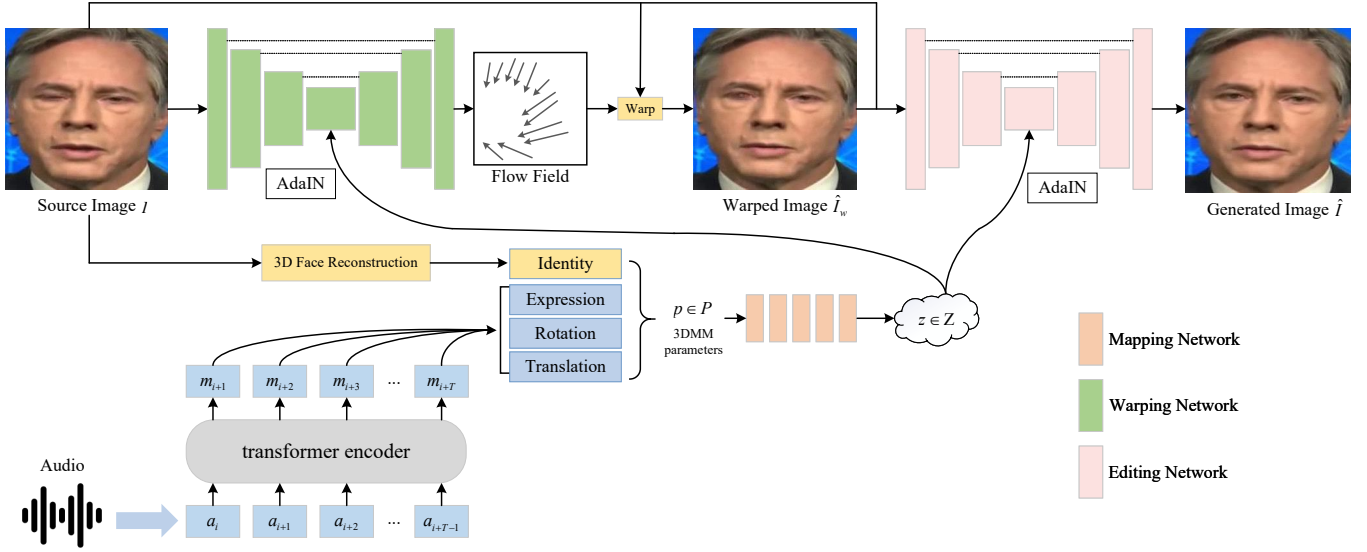


Figure 1: Overview of our framework. Given input audio, the transformer-based audio encoder predicts the corresponding parameters of 3DMM. These parameters concatenated with identity parameters are fed into a mapping network to obtain a latent code z . Then a warped image is first generated from the source image and further edited to the final image.

In general, this work’s contributions are summarized as follows:

- We propose a transformer-based audio encoder to capture long-term context from audio sequences. This audio encoder can help eliminate the ambiguities from the audio feature and improve the lip-synchronization.
- We propose to take usage of the identity parameters of 3DMM, which can help learn each speaker’s style and bring improvement for lip-movement accuracy and generated image quality.
- Experimental results on talking head generation track demonstrate that our method outperforms other competition methods in 4 of 11 evaluation metrics and ranks first in this task.

2 RELATED WORK

Most of the talking head generation methods [6, 12, 17, 22, 27, 28, 30, 32] are audio-driven, trying to generate the videos in sync with the input audio. Prajwal *et al.* [17] proposed a lip-sync discriminator to enhance the lip-sync between generated video and input audio. Guo *et al.* [6], Liu *et al.* [12] and Yao *et al.* [27] tried to generate the high-fidelity talking head video based on the Neural Radiance Field (NeRF). Zhang *et al.* [30] designed a module to encode the contextual information with the phonetic information of each individual frame. In our work, we use transformer to extract the long-term contextual information of audio sequences.

Since it is challenging to generate the lip-sync talking head video according to the audio signal directly, several works [9, 11, 14, 19, 25, 26, 31] take use of 2D landmarks or 3D face models as intermediate representations. Xie *et al.* [26] proposed a two-stage framework to deal with the heterogeneous data by taking facial landmarks as intermediate prior of latent representations. Yi *et al.* [29] and Zhang *et al.* [30] took the reconstructed 3D facial image as explicitly intermediate representation and re-render it into synthesized head

image. In our work, we use the parameters of 3DMM implicitly without reconstructing the 3D facial image.

3 METHODOLOGY

3.1 Problem Formulation

Given an input audio A and a source image I of the speaker, we aim to generate a head video V whose lip movement is in sync with the input audio A . As shown in Fig. 1, we first map the input audio A into expression, rotation and translation parameters $E(A)$ of 3DMM [16]. Then we extract the identity parameters of the source image and generate the talking head video V based on all of the parameters p and source image I .

3.2 Transformer-based Audio Encoder

The transformer-based audio encoder E aims to synthesize expression, rotation and translation parameters $m = \{\beta, R, t\}$ conditioned on audio feature $A = \{a_i, a_{i+1}, \dots, a_{i+T-1}\}$ with T acoustic feature, where a_i denotes the acoustic feature corresponding to the i -th frame I_i . We first add positional encoding to A and then linearly transform it into value matrix V , query matrix Q and key matrix K . Then the attention mechanism is applied as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V, \quad (1)$$

where d denotes the dimension of query and key vector.

Moreover, we utilize the multi-head attention to enhance the representation ability of model. The output is the concatenation of h attention head.

$$\text{MultiHead}(Q, K, V) = \oplus(H_1, H_2, \dots, H_h)W_{out}, \quad (2)$$

where $H_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$, W_i^Q, W_i^K, W_i^V and W_{out} are the projection matrix. Similar to transformer encoder in [21],

residual connection followed by layer normalization and position-wise fully connected feed-forward network are used. Our transformer-based audio encoder includes a stack of $N = 4$ identical layers consisting of multi-head self-attention mechanism, position-wise fully connected feed-forward network. Finally, a fully connected layer is used to transform the hidden state into predicted parameters m . Since there always exists a time delay between the lip/expression movement and audio in the actual speaker system [1][2], the prediction of audio encoder is one frame ahead of the input audio.

To train the audio encoder, we measure the euclidean distance between prediction m and ground truth m^{gt} obtained from the 3D face reconstruction network [5] as:

$$L_{MSE} = \frac{1}{T} \sum_{j=1}^T \|m_{i+j} - m_{i+j}^{gt}\|_2. \quad (3)$$

3.3 Implicit Learning of 3D Face

Based on the 3DMM, the 3D face shape S is parameterized as a model of facial expression and facial identity:

$$S = \bar{S} + \alpha B_{id} + \beta B_{exp}, \quad (4)$$

where $\bar{S}$ is the average face shape, B_{id} and B_{exp} are the bases of identity and expression computed via Principal Component Analysis (PCA). $\alpha \in \mathbb{R}^{80}$ and $\beta \in \mathbb{R}^{64}$ are the identity and expression parameters respectively. According to Eq. (4), the 3D face shape has close correlation with identity parameters and expression parameters. What's more, each speaker speaks the same sentences differently because of his/her own style and identity parameters can help reflect this feature. The model can learn the style of each speaker by concatenating the identity parameters α with $E(A)$.

3.4 Rendering Network

With parameters $p = E(A) \oplus \alpha$ and image I as input, we follow Ren *et al.*'s method [18] to generate the final photo-realistic images. We first map parameters p to a latent vector z . Then a coarse image $\hat{I}_w$ is generated by warping source image I according to latent vector z . Then an editing network takes the coarse image and source image as input and generates the final image $\hat{I}$ according to latent vector z again. The latent vector z is injected to the warping network and editing network by AdaIN [8] operations. The training loss is formulated as

$$L = \lambda_w \sum_i \|\phi_i(I_t) - \phi_i(\hat{I}_w)\|_1 + \lambda_c \sum_i \|\phi_i(I_t) - \phi_i(\hat{I})\|_1 + \lambda_s \sum_j \|G_j^\phi(I_t) - G_j^\phi(\hat{I})\|_1, \quad (5)$$

where I_t denotes the target image, ϕ_i is the activation map of the i -th layer of VGG-19 network, G_j^ϕ is the Gram matrix constructed from activation maps ϕ_j , λ_w , λ_c and λ_s are the weighted factors.

4 EXPERIMENTS

4.1 Datasets and Evaluation Metrics

Datasets. The ViCo dataset contains the speaker videos along with the paired listener videos. In this paper, we only take use of the speaker videos as training set. In this challenge, we use 325 speaker videos with resolution of 256×256 and 30 FPS.

Evaluation Metrics.

Generation quality (image level): The perceptual-based no-reference objective image sharpness metric (CPBD) [15] can evaluate the sharpness of the generated images without target images. And the Peak Signal to Noise Ratio (PSNR) and Structure Similarity Index Measure (SSIM) [24] can evaluate the quality of generated images by comparing with target images.

Generation quality (feature level): Fréchet Inception Distance (FID) [7] evaluates the similarity of the generated images and target ones in feature level.

Identity preserving: Identity preserving is evaluated by extracting identity features predicted by ArcFace [4] and calculate the cosine similarity (CSIM).

Expression: Expression is evaluated by calculating the L1 distance of 3DMM expression features (ExpFD).

Head motion: Head motion is evaluated by calculating the L1 distance of 3DMM angle (AngleFD) and translation features (TransFD).

Lip sync: The synchronization between lip movement and audio is evaluated by the AV offset (AVOffset) and AV confidence (AVConf) predicted by SyncNet [3].

Lip landmark distance: We evaluate the accuracy of lip-sync by calculating the L1 distance of lip landmarks (LipLMD).

4.2 Implementation Details

The training scheme of our framework consists of two stage in ViCo dataset. We first train audio encoder with a batch size of 512 for 2000 epochs. AdamW optimizer [13] is used with learning rate of 10^{-4} and weight decay of 0.01. We employ $h = 4$ heads and 96-dimensional hidden state in attention mechanism and use $T = 90$ acoustic feature. Then we finetune the rendering network based on the pretrain model of [18] for 400 epochs with minor change to the first layers of mapping network. The parameters λ_w , λ_c and λ_s in Eq. (5) are equal to 2.5, 4 and 1000, respectively.

4.3 Quantitative Evaluation

As shown in Tab. 1, our method ranks first in the talking head generation track by ranking first in 4 metrics (PSNR, CSIM, ExpFD and LipLMD) of all 11 evaluation metrics. Our method achieves the first place in SSIM and PSNR, the third place in CPBD and FID, which indicates that our method can generate more high-quality images than other competitors' methods.

4.4 Qualitative Visualizations

We show the visualization results of our model in Fig. 2. The results are from the validation set since we have no access to the test set. In Fig. 2, our method first generates a warped image based on the random input source image. And then the warped image is edited with more details to eliminate the artifacts. We can see that our method can generate high-quality images both in visualization and lip-movement accuracy.

4.5 Ablation Studies

The result of ablation studies is shown in Tab. 2. The baseline model provided by the organizers uses an LSTM-based encoder to extract audio feature and does not incorporate identity parameters. Baseline + ID denotes that the baseline model incorporates identity

Table 1: Quantitative comparisons of other teams’ methods and our method. Better results are highlighted in bold.

Rank	Team Name	SSIM	CPBD	PSNR	FID	CSIM	ExpFD	AngleFD	TransFD	AVOffset	AVConf	LipLMD
1	sysu_hcp	0.608	0.175	17.767	29.709	0.144	0.131	0.056	0.057	2.167	0.867	10.101
2	Megvii_goodjuice	0.574	0.218	17.179	24.678	0.124	0.137	0.056	0.056	1.200	3.153	10.646
3	iLearn	0.565	0.134	16.546	25.050	0.135	0.137	0.063	0.064	1.000	8.045	10.900
3	Avatar	0.608	0.163	17.696	34.571	0.125	0.140	0.058	0.062	4.433	0.626	11.643
3	Digital_Human	0.595	0.145	17.331	30.944	0.130	0.159	0.067	0.073	0.233	0.578	12.837
3	THU-Talking	0.559	0.402	16.390	45.361	0.123	0.168	0.062	0.071	2.400	1.234	12.707

Table 2: Ablation experiments.

Method	SSIM	CPBD	PSNR	FID	CSIM	ExpFD	AngleFD	TransFD	AVOffset	AVConf	LipLMD
Baseline	0.552	0.146	15.913	32.424	0.135	0.162	0.071	0.064	0.700	3.601	12.086
Baseline + ID	0.566	0.176	16.499	28.143	0.139	0.138	0.066	0.059	0.467	2.447	11.809
Baseline + transformer	0.595	0.148	17.071	34.266	0.145	0.156	0.061	0.062	3.867	1.089	10.297
Ours	0.608	0.175	17.767	29.709	0.144	0.131	0.056	0.057	2.167	0.867	10.101

**Figure 2: Visualization results.**

parameters while using LSTM-based encoder and Baseline + transformer denotes that the baseline model uses transformer-based audio encoder instead of LSTM-based encoder while not incorporating identity parameters.

Compared with baseline model, incorporating identity parameters can improve the sharpness of generated images reflected by improvement in CPBD and using transformer-based audio encoder can improve the accuracy of lip-movement and maintain the identity of speakers.

5 CONCLUSIONS

In this work, we tackle the talking head generation task by proposing a transformer-based audio encoder and exploring the importance of 3DMM’s identity parameters. The transformer-based audio

encoder can capture the long-term context from audio, improving the accuracy of lip-movement. And the identity parameters can help improving the image quality. The experiments show that our method ranks first in the talking head generation challenge by ranking first in 4 of all 11 evaluation metrics.

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